

	$\hat{\beta}$
Constant	0.3317
Trend	0.2668

Fisher reported: 0.2668.

The slope is 0.2668: each passing year, the yield advantage of nitrate of soda over sulphate of ammonia grew by about 0.27 bushels per acre. Over the thirty year span, that accumulates to roughly eight bushels. Fisher's question was whether one nitrogen source was gaining an advantage, and this trend suggests it was.

8.2.1 Multicollinearity

You may recall one of our most basic requirements in the simple linear regression formula for $\hat{\beta}_1$ was *sample variation in X* : we needed $\sum_{i=1}^n (X_i - \bar{X})^2 \neq 0$, because that sum sits in the denominator of $\hat{\beta}_1$. We cannot divide by zero. (And practically speaking, if X never varies, why would we be looking for a relationship between X and Y in the first place?)

In multiple regression with explanatory variables $X = x_1, \dots, x_k$, the same idea generalizes: we need variation not just in each X_j individually, but also in their linear combinations. The last step in the derivation of Equation 8.3 requires $(X'X)$ to be invertible, which means X must have full column rank. This is the generalization of “variation in X ” to the case of many explanatory variables.

Classical Linear Regression Assumption 4: No Perfect Multicollinearity

The $n \times k$ design matrix X has full column rank: $\text{rank}(X) = k$: No column of X can be written as a linear combination of the other columns.

To see why, consider a simple example. Suppose your model has an intercept and one X variable, but that variable never changes:

$$X = \begin{pmatrix} 1 & 5 \\ 1 & 5 \\ 1 & 5 \end{pmatrix}.$$

Column 2 is just 5 times Column 1. The columns do not contain independent information: X does not have full column rank. As a result, $X'X$ is singular and cannot be inverted, so OLS cannot produce estimates. This is **perfect multicollinearity**: one column is an exact linear combination of another.

In ordinary algebra, you can divide by a number only if it is not zero. If you try to divide by zero, the operation is undefined. In matrix algebra, “dividing by” $X'X$ means multiplying by $(X'X)^{-1}$. That operation is only possible if the inverse exists. When

the columns of X are linearly dependent, $X'X$ has no inverse.⁶ In practice, statistical software detects this problem and drops one of the perfectly collinear columns so that the remaining variables contain independent information and the model can be estimated.

8.2.2 Unbiasedness of OLS

In assessing the quality of our model, we should ask a simple question: is it biased? In other words, if we could repeat this estimation process over and over again on new samples drawn from the same population, would our estimates tend to land on the true parameter values, or would they systematically miss? *Systematically* is the key word. An estimate can be far from the truth in a particular sample and still be unbiased. Bias is not about being wrong once. It is about being wrong in a predictable direction on average. So when we ask whether OLS is unbiased, we are asking whether its statistical *expectation* equals the true parameter value.

We can make a quick check that OLS, if our assumptions hold, is unbiased. Given $\hat{\beta} = (X'X)^{-1}X'y$ and $y = X\beta + u$, we can substitute $X\beta + u$ in for y . Following some quick algebra, we can see that OLS is indeed unbiased:

$$\begin{aligned}
 \hat{\beta} &= (X'X)^{-1}X'y && \text{OLS formula} \\
 &= (X'X)^{-1}X'(X\beta + u) && \text{Substitute } y = X\beta + u \\
 &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'u && \text{Distribute} \\
 &= \beta + (X'X)^{-1}X'u && \text{Since } (X'X)^{-1}X'X = I_k \quad (8.4)
 \end{aligned}$$

We see that our estimate, $\hat{\beta}$, equals the true parameter β plus the term $(X'X)^{-1}X'u$. So if there is bias, it must come from that extra piece. Bias, by definition, is a systematic error. To determine whether OLS is biased, we take expectations:

⁶ In the simple regression case, this reduces to the requirement that x must vary. If every x_i is the same, then the second column of X is just a constant multiple of the intercept column. There is no independent movement in x , so we cannot separate its effect from the intercept. Algebraically, $\sum (x_i - \bar{x})^2 = 0$, which makes $S_{xx} = 0$ and prevents us from dividing by it. The matrix condition that $X'X$ be invertible is the general version of the requirement that we not divide by zero.